

A Pedagogical Introduction to Equalizers, Kernels, Limits, and Colimits

Compiled from Category Theory Notes

Contents

1	Motivation: Solving Equations in Sets	1
2	Equalizers	2
2.1	Definition via Universal Property	2
2.2	Examples	2
3	Kernels as Special Equalizers	2
4	Coequalizers and Cokernels	3
4.1	Coequalizer	3
4.2	Cokernel	3
5	The Big Picture: Limits and Colimits	4
5.1	Cones and Limits	4
5.2	Cocones and Colimits	5
6	Summary Table	5

1 Motivation: Solving Equations in Sets

In set theory, we often encounter the problem of finding elements where two functions agree. Given $f, g : A \rightarrow B$, the *solution set* is

$$\{x \in A \mid f(x) = g(x)\}.$$

This subset comes with an inclusion map $i : \{x \in A \mid f(x) = g(x)\} \hookrightarrow A$, and it has a special property: any function $h : Z \rightarrow A$ that satisfies $f \circ h = g \circ h$ must factor uniquely through i . Category theory abstracts this idea into the concept of an **equalizer**.

2 Equalizers

2.1 Definition via Universal Property

Let $\mathcal{C}$ be a category and let $f, g : X \rightarrow Y$ be two parallel morphisms.

Definition 2.1. An **equalizer** of f and g is a pair (E, eq) consisting of an object E and a morphism $\text{eq} : E \rightarrow X$ such that:

1. $f \circ \text{eq} = g \circ \text{eq}$,
2. For any object Z and morphism $m : Z \rightarrow X$ with $f \circ m = g \circ m$, there exists a *unique* morphism $u : Z \rightarrow E$ making the following diagram commute:

$$\begin{array}{ccccc} Z & & & & \\ \downarrow \exists! u & \searrow m & & & \\ E & \xrightarrow{\text{eq}} & X & \xrightleftharpoons[f]{g} & Y \end{array}$$

The morphism eq is necessarily a **monomorphism** (i.e. satisfying left elimination, it is injective in many concrete categories). The equalizer is unique up to unique isomorphism.

2.2 Examples

Example 2.1 (In Set). For sets X, Y and functions $f, g : X \rightarrow Y$, the equalizer object is

$$E = \{x \in X \mid f(x) = g(x)\},$$

and $\text{eq} : E \hookrightarrow X$ is the subset inclusion. The universal property says: any function $m : Z \rightarrow X$ whose image lies in E factors uniquely through E .

Example 2.2 (In Grp). Given group homomorphisms $f, g : G \rightarrow H$, the equalizer is the subgroup

$$E = \{x \in G \mid f(x) = g(x)\},$$

with inclusion homomorphism. This is precisely the kernel of the difference map $f - g$ when H is abelian.

3 Kernels as Special Equalizers

In categories equipped with a **zero object** (and thus zero morphisms $0_{X,Y} : X \rightarrow Y$), we can specialize the equalizer construction.

Definition 3.1. Let $\mathcal{C}$ be a category with zero morphisms. The **kernel** of a morphism $f : X \rightarrow Y$ is the equalizer of f and the zero morphism $0_{X,Y} : X \rightarrow Y$:

$$\text{Ker}(f) = \text{Eq}(f, 0_{X,Y}).$$

Diagrammatically:

$$\text{Ker}(f) \xrightarrow{\text{ker}} X \xrightarrow[0]{f} Y$$

Universality of the kernel:

$$\begin{array}{ccc} Z & & \\ \downarrow \exists! u & \searrow m & \\ \text{Ker}(f) & \xrightarrow{\text{ker}} & X \xrightarrow[0]{f} Y \end{array}$$

Example 3.1 (In $\mathbf{Vect}_k$). For a linear map $T : V \rightarrow W$, the kernel is the subspace $\{v \in V \mid T(v) = 0\}$. The inclusion map is the equalizer.

4 Coequalizers and Cokernels

By **reversing all arrows**, we obtain the dual notions.

4.1 Coequalizer

Definition 4.1. A **coequalizer** of $f, g : X \rightarrow Y$ is a pair (Q, q) where $q : Y \rightarrow Q$ satisfies $q \circ f = q \circ g$, and for any $(Z, h : Y \rightarrow Z)$ with $h \circ f = h \circ g$, there exists a unique $v : Q \rightarrow Z$ making the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow[g]{f} & Y \xrightarrow{q} Q \\ & \searrow h & \downarrow \exists! v \\ & & Z \end{array}$$

In **Set**, the coequalizer of $f, g : X \rightarrow Y$ is the quotient set $Y/\sim$, where $\sim$ is the equivalence relation generated by $f(x) \sim g(x)$ for all $x \in X$.

4.2 Cokernel

Definition 4.2. In a category with zero morphisms, the **cokernel** of $f : X \rightarrow Y$ is the coequalizer of f and $0_{X,Y}$:

$$\text{Coker}(f) = \text{Coeq}(f, 0_{X,Y}).$$

Diagram:

$$X \xrightarrow[0]{f} Y \xrightarrow{\text{coker}} \text{Coker}(f)$$

Universality:

$$\begin{array}{ccc} X & \xrightarrow[0]{f} & Y \xrightarrow{\text{coker}} \text{Coker}(f) \\ & \searrow h & \downarrow \exists! v \\ & & Z \end{array}$$

Example 4.1 (In $\mathbf{Vect}_k$). The cokernel of a linear map $T : V \rightarrow W$ is the quotient space $W/\text{Im}(T)$. The projection map is the coequalizer.

5 The Big Picture: Limits and Colimits

Both equalizers and kernels are instances of a far-reaching concept: **limits**. Dually, coequalizers and cokernels are **colimits**.

5.1 Cones and Limits

Let J be a small “index” category (think of it as a pattern), and let $D : J \rightarrow \mathcal{C}$ be a **diagram** of shape J in $\mathcal{C}$.

Definition 5.1. A **cone** over D consists of an object L (the apex) and a family of morphisms $\pi_X : L \rightarrow D(X)$ for every object X in J , such that for every morphism $u : X \rightarrow Y$ in J , the following triangle commutes:

$$\begin{array}{ccc} & L & \\ \pi_X \swarrow & & \searrow \pi_Y \\ D(X) & \xrightarrow{D(u)} & D(Y) \end{array}$$

A **limit** of D , denoted $\lim D$, is a **universal cone**: a cone (L, π) such that for any other cone (Z, τ) over D , there exists a *unique* morphism $u : Z \rightarrow L$ with $\tau_X = \pi_X \circ u$ for all X .

$$\begin{array}{ccccc} Z & & \xrightarrow{\tau_Y} & & D(Y) \\ & \searrow \exists! u & & \nearrow \pi_Y & \\ & \lim D & & & \\ & \downarrow \pi_X & & \nearrow D(u) & \\ & D(X) & & & \end{array}$$

τ_X (curved arrow from Z to $D(X)$)

Example 5.1 (Equalizer as a Limit). The equalizer of $f, g : X \rightarrow Y$ is the limit of the diagram

$$X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} Y$$

The index category J has two objects and two parallel non-identity arrows. The limit cone picks out the subobject E with its map to X (other trivial arrows are usually not drawn).

Example 5.2 (Product as a Limit). If J is a discrete category with two objects and no non-identity arrows, a diagram $D : J \rightarrow \mathcal{C}$ is just a pair of objects A, B . The limit is the product $A \times B$.

5.2 Cocones and Colimits

Dualizing everything yields colimits.

Definition 5.2. A **cocone** under D consists of an object C and morphisms $\iota_X : D(X) \rightarrow C$ such that for every $u : X \rightarrow Y$ in J , the following commutes:

$$\begin{array}{ccc} D(X) & \xrightarrow{D(u)} & D(Y) \\ & \searrow \iota_X & \swarrow \iota_Y \\ & C & \end{array}$$

The **colimit** of D , denoted $\text{colim } D$, is a **universal cocone**: a cocone (C, ι) such that for any other cocone (Z, τ) , there is a unique $v : C \rightarrow Z$ with $\tau_X = v \circ \iota_X$.

$$\begin{array}{ccc} D(X) & \xrightarrow{D(u)} & D(Y) \\ & \searrow \iota_X & \swarrow \iota_Y \\ & \text{colim } D & \\ & \downarrow \exists! v & \\ & Z & \end{array}$$

τ_X (curved arrow from $D(X)$ to Z) τ_Y (curved arrow from $D(Y)$ to Z)

Example 5.3 (Coequalizer as a Colimit). The coequalizer of $f, g : X \rightarrow Y$ is the colimit of the parallel-arrow diagram. The universal cocone gives the quotient object.

Example 5.4 (Coproduct as a Colimit). For a discrete diagram $\{A, B\}$, the colimit is the coproduct $A \sqcup B$.

6 Summary Table

Construction	Limit or Colimit?	Diagram Shape
Equalizer	Limit	$\bullet \rightrightarrows \bullet$
Kernel	Limit	$\bullet \rightrightarrows \bullet$ (one arrow zero)
Product	Limit	$\bullet \quad \bullet$ (discrete)
Pullback	Limit	$\bullet \rightarrow \bullet \leftarrow \bullet$
Coequalizer	Colimit	$\bullet \rightrightarrows \bullet$
Cokernel	Colimit	$\bullet \rightrightarrows \bullet$ (one arrow zero)
Coproduct	Colimit	$\bullet \quad \bullet$ (discrete)
Pushout	Colimit	$\bullet \leftarrow \bullet \rightarrow \bullet$

Final Remarks

The language of limits and colimits unifies a vast array of constructions across algebra, topology, and logic. Mastering equalizers and coequalizers is the first step toward understanding this elegant framework.